



Items and Inventory: **Use Weights as Units of Measure**

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the topic of using weights as units of measure.

Objectives



At the end of this topic, you will be able to:

- Set up weights in unit of measure groups

In this topic, you will learn how to set up weights as a unit of measure in a unit of measure group. To fully understand this topic, you need to already know how to set up unit of measure groups.

Business Scenario



OEC Computers buys cables for resale. Previously the cables were purchased by the meter. However they are changing to a new supplier who sells the cables by weight.

They would like to add a weight unit of measure to the existing unit of measure group. Then set that measure as the default measure for purchasing.

They would also like to create a new unit of measure group for USB connectors which uses weight as a base unit of measure.

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Business Example 1 – Adding a weight unit of measure to a group



OEC is changing suppliers for cables.

The new supplier sells cables by weight.

OEC will purchase cables in kilograms (kg) and manage them in inventory by meters.

- 1 meter Network cable = 1 kg
- 1 meter of HDMI cable = 2 kg
- 1 meter of Power cable = 4 kg

We will change an existing unit of measure group to support purchasing these items by weight while selling the items in meters.

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- OEC computers resells different types of cables. A unit of measure group already exists for this item with meter as the base unit of measure.
- Previously the cables were purchased in 100 meter reels. OEC has decided to change to a new supplier who sells the cables by weight.
- OEC will purchase cables by weight and manage them in inventory by meters. These cables will be sold in various units that convert to meters.
- Because cables are purchased by weight and sold by lengths in meters, the unit of measure group of cables must be able to convert quantities in kilograms into quantities in meters
- Here are 3 examples of cable items managed in OEC Computers : Network cable, HDMI cable and Power cable.
- A **meter** inventory UoM is defined for each item. Each **meter** unit has a different weight:
 - 1 **meter** of **network cable** weighs **1 kg**,
 - 1 **meter** of **HDMI cable** weighs **2 kg** and
 - 1 **meter** of **industrial power cable** weighs **4 kg**.
- We will look at how we can change an existing unit of measure group to support purchasing these items by weight while storing and selling the items in meters.

Using Weight in a Unit of Measure Group

Select the existing group and choose *Group Definition*.

Unit of Measure Groups

#	Group	Group Description
1	Manual	Manual
2	Paper	Paper
3	Cable	Cable

Group Definition – Cable - Setup

#	Alt Qty	Alt UoM	Weight Factor	=	Base Qty	Base UoM
1	1	Meter		=	1	Meter
2	1	Reel		=	100	Meter
3	1	Spool		=	50	Meter
4	1	Kilo	kg	=	1	Meter

Enter the weight unit in the *Alternate Unit of Measure* field.

Use a weight factor when you need to adjust the unit quantity based on the weight of the item.

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- In our business example, we currently have a unit of measure group for cables.
- We would like to add kilo to the unit of measure group so that we can use it when purchasing the cables.
- First we define kilo as a unit of measure and set it equal to a weight of 1 kilogram in the *Units of Measure - Setup* window.
- Then we open the *Unit of Measure Group* window to enter our new unit to the cable unit of measure group.
- We select the group and choose the *Group Definition* button to open a window to define the relationship between unit of measure codes within the group.
- In the existing group, Meter is already set as the base unit and all other units in the same group are related to the base unit by conversion rules.
- Since we purchase the cables by weight in kilograms, we enter the unit Kilo as an alternative unit of measure.
- Then we need to enter the fields alternative quantity and base quantity. In our example, 1 kilogram is equal to 1 Meter of cable for our most popular types of cables.
- However, some cable items have a different weight, therefore we also enter a weight factor. The factor will allow us to easily adjust the unit based on the weight of the item.

Advantage of Weight Factor

Network Cables 1 meter = 1 kg
HDMI Cables 1 meter = 2 kg
Power cables 1 meter = 3 kg

Purchasing Data Sales Data Inventory Data

Method By Warehouse

2kg

Use the weight specified in the item master record for the conversion.

Group Definition – Cable - Setup

#	Alt Qty	Alt UoM	Weight Factor	=	Base Qty	Base UoM
1	1	Meter		=	1	Meter
4	1	Kilo	kg	=	1	Meter

"Alt Qty" x "Weight Factor" (from Item Master Weight) = Base Qty in Document

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Let's talk for a moment of the advantage of using a weight factor in situations where we want to convert quantities in weights to quantities in lengths.

In previous releases, we did not have a weight factor and it was much more difficult to manage items that had different weights while having the same length, width or height. Now with a weight factor we can use the weight listed in the item master record to make the conversion. This makes it possible to share the same unit of measure group for cable rather than having three separate unit of measure groups.

For example, in our company we have 3 general types of cable: network cables, HDMI cables and power cables and they all have different weights. For network cables, 1 meter equals 1 kilogram. HDMI cables are twice as heavy so 1 meter equals 2 kilograms. Power cables are heavier still, with 3 kilograms for each meter.

Now we can use the weight factor field to calculate the weight different for the meter unit of measure. Because we have entered the kilogram weight factor, the system will look at the weight specified in the item master and calculate the actual weight being purchased, sold or inventoried.

The system multiplies the alternative quantity by the weight from the item master to calculate the amount of base quantity for a document.

Example of how conversion works

- The weight factor value comes from the item master weight field.
- If the unit differs, a calculation is performed.
For example: converting grams to kilograms.
- Then calculations are made to convert the Alternative Quantity (Kilo) to the Base Quantity (Meter).

- The Weight factor of the network cable = 1 kg
- The Weight factor of the HDMI cable = 2 kg
- The Weight factor of the power cable = 4 kg



- 1 Kg of the network cable = 1 meter
- 1 Kg of the HDMI cable = 0.5 meter
- 1 Kg of the power cable = 0.25 meter

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- Let's look at how this works with our business example.
- The value for the weight factor comes from the value in the Item Master Data *Weight* field.
- If the weight of the item was entered in a different weight measurement than the one indicated in the factor field, then the system converts the weight according to the weight factor. For example, if the weight of the item is 2000 grams then the factor will be 2 (kilograms).
- We have set conversion rules in our unit of measure group. So calculations are performed to convert the alternative quantity (in our case in kilograms) to the base quantity (in our case meters). Remember the formula Alternative Quantity multiplied by the weight factor equals the base quantity. Logically this also means that an alternative quantity of a unit equals the base quantity of the unit divided by the weight factor.
- Therefore the following is true for our cable items.
- Since we manage the cable inventory by kilograms, we entered the following weights in the Weight field of the *Item Master data*:
 - 1 kg for the network cable
 - 2 kg for the HDMI cable and
 - 4 kg for the power cable
- This means that:
 - 1 kilogram unit of the network cable equals 1 meter unit
 - 1 kilogram unit of the HDMI cable equals 0.5 meter unit
 - 1 kilogram unit of the power cable equals 0.25 meter unit

Effect of Weight Factor in Documents

Delivery

Customer: C23900
 Name: Parameter Technology
 Contact Person: Daniel Brown
 Customer Ref. No.:
 BP Currency: GBP

- The Weight factor of the network cable = 1 kg
- The Weight factor of the HDMI cable = 2 kg
- The Weight factor of the power cable = 4 kg

Contents Logistics Accounting

#	Item No.	Item Description	Quantity	UoM Name	Items per Unit	Unit Price	Total (LC)
1	C10001	Network Cable	1	Kilo	1.000	GBP 2.00	GBP 2.
2	C10002	HDMI Cable	1	Kilo	0.500		
3	C10003	PowerCable	1	Kilo	0.250		
4					0.000		

- 1 Kg of the network cable = 1 meter
- 1 Kg of the HDMI cable = 0.5 meter
- 1 Kg of the power cable = 0.25 meter

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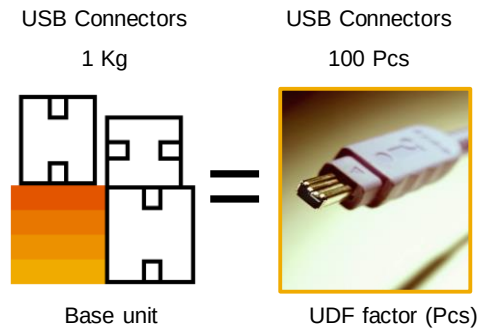
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Once we have made the settings, we test the items and units of measure in a document.

Notice how the calculation using the weight factor affects the Item per Unit field. For example, look at the power cable item. Since the weight factor of the power cable is 4 kilograms. If we purchase one kilogram worth of power cable, the length is only a quarter of a meter long.

Business Example 2: Weight as the Base unit

- OEC Computers resells USB connectors.
- The USB connectors are bought and stored in kilograms but sold as pieces.
- Therefore it is a good practice to use a weight as the base unit.
- We can create a user-defined field to specify the conversion between the base unit in weights and another type of unit, such as pieces.
- Then we can use the *UDF Factor* field in the *Unit of Measure Group* window to handle the conversion.



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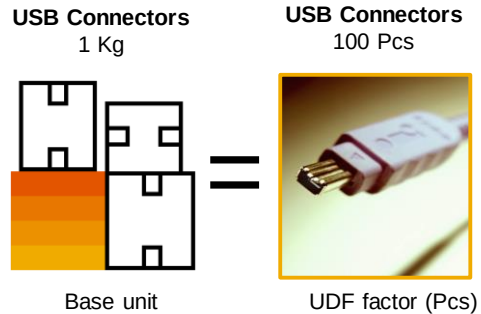
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- In our previous business example, the cables are stored in meters so we have chosen meters as the base unit. For other items, a weight may need to be a base unit.
- For example, OEC Computers resells USB connectors. The USB connectors are bought and stored in kilograms. However they are sold individually as pieces.
- The inventory unit is a weight (kilograms). Therefore, it is a good practice to use the weight as the base unit. Since the item is sold with a non-weight unit, we need to convert from a weight to a non-weight unit.
- To do this, we can create a user-defined field to specify the conversion between the base unit in weights and another type of unit, such as pieces.
- Then we can use the *UDF Factor* field in the *Unit of Measure* group to handle the conversion.

Steps for using a UDF Field

Group Definition – Connectors - Setup						
#	Alt Qty	Alt UoM	UDF Factor	Weight Factor =	Base Qty	Base UoM
1	1	Kg		=	1	Kg
2	1	Pcs	Pcs	=	1	Kg

- Set up a user-defined field (UDF) in the item master
- The user-defined field contains conversion value (e.g. 100)
- Enter the UDF name in the UDF factor field
- Calculates conversion of 100 pcs for each 1 kg of connectors



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- To use the UDF Field take the following steps: Set up a user defined field in the item master. The user-defined field contains conversion values from the base unit in weight to alternative units.
- Let's look at an example. OEC computers sells connectors. Connectors are managed by weight in the inventory. The base and inventory units are set to Kilograms. In the graphic, we can see the Connectors unit of measure group. The base unit is set to Kilogram and the alternative unit is pieces.
- One of the items using this unit of measure group is the USB connector item. A user-defined field was added to the item to indicate how many connector pieces are equal to 1 kilogram of connectors. Then the user-defined field name was entered into the UDF Factor field.
- The UDF field value of the USB connector item is 100. This means that 100 pieces equal 1 kg of connectors.
- Notice that the weight factor is not needed because weight is the base unit.
- A UDF factor can also be helpful when you want to record additional measurements for an item such as density or volume or any other element needed for conversion rules.

Example 3: Use UDF to Convert Length to Volume

Use a UDF field to convert between types of measurements such as length to volume.

Group Definition – Box - Setup						
#	Alt Qty	Alt UoM	UDF Factor	Weight Factor =	Base Qty	Base UoM
1	1	Inch		=	1	Inch
2	2	Liter	Liters	=	6	Inch



The Liters UDF factor of a box item = 2 liters



2 Liters of the box item = 6 inches

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- A UDF factor can also be helpful when you want to convert between different types of measurements. For example if you wish to record additional measurements for an item such as density or volume or any other element needed for conversion rules.
- OEC Computers keeps a supply of boxes that are used as packaging materials. These boxes are managed as items and tracked in the inventory.
- When boxes are folded they are flat and measured in inches. The boxes can also be measured by their volume, in our case we use liters.
- In the graphic, we see the Box unit of measure group that has been created for these items.
- A *UDF Factor* field has been defined for liters and has been assigned in the *Item Master Data* header for the box items.
- For the most commonly used box item, the value of the Liters UDF field is 2. This means that 2 liters of the box item equals 6 inches.

Summary



Here are some key points:

- Units based on weight can be used in units of measure groups.
- A weight factor can be used to convert units of weight to a base unit.
- The weight factor value is derived from the *Weight* field in the Item Master record.
- The formula for calculating weight is:
 $\text{Alt Qty} \times \text{Weight Factor (from Item Master Weight)} = \text{Base Qty}$
- SAP Business One has provided a UDF Factor field in the Unit of Measure Group that allows you to include values from a user-defined field to perform additional conversions for units of measure.
- The UDF factor allows you to have a weight unit as the base unit of measure in a group.

- A weight factor can be used to convert units of weight to the base unit.
- The weight factor value is derived from the Weight field in the Item Master record.
- When using a weight factor, the formula for the calculating quantities is:
“Alternative Quantity” multiplied by the “Weight Factor “(from Item Master Weight)
equals the Base Quantity.
- Additionally Business One has provided a UDF Factor field in the Unit of Measure Group window that allows include values from a user-defined field in the item master to perform additional conversions for units of measure. For example, a UDF could be used to convert lengths or widths to densities or volumes.
- The UDF factor also allows you to have a weight unit as the base unit of measure in a group.

Thank you

You have completed the topic on using weights as units of measure.

Thank you for your time!

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